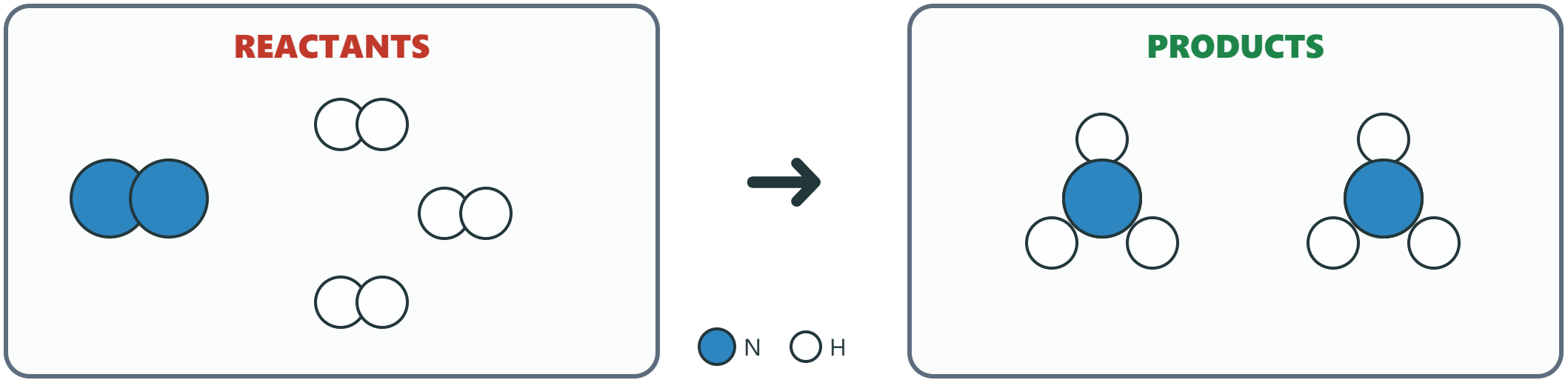
Atoms are not created or destroyed: they are rearranged



$N_2 + 3\ H_2$ → $2\ NH_3$

Reactants: **2 N** + **6 H**

Products: **2 N** + **6 H** ✓

⚖️ **Law of conservation of mass** (Lavoisier): mass of reactants = mass of products · 28 g N_2 + 6 g H_2 = 34 g NH_3